

## MAJOR PROJECT REGISTRATION FORM

**Date:**

|                    |                                            |                                                 |           |         |
|--------------------|--------------------------------------------|-------------------------------------------------|-----------|---------|
| Project Category:  | <input type="checkbox"/> Research oriented | <input type="checkbox"/> Application oriented   | Batch ID: | AIML-A3 |
|                    | <input type="checkbox"/> Product oriented  | <input type="checkbox"/> Other (please specify) |           |         |
| Specify if others: |                                            |                                                 |           |         |

**Title of the Project:** Explainable Multimodal Foundation Model for Early Breast Cancer Diagnosis through Cross-Modal Fusion of Mammography, Ultrasound, Clinical Metadata, and Vision Foundation Models.

### Abstract

Radiologists screening for breast cancer frequently encounter reduced diagnostic sensitivity when relying on a single imaging modality, particularly in patients with dense breast tissue, where mammography alone yields elevated false-negative and false-positive rates and where diagnostically relevant information from complementary ultrasound scans and structured clinical risk factors, such as age, breast density, family history, and palpation findings, is not systematically combined into the decision process. This gap is most acute for radiologists and sonologists working in high-volume community and tertiary breast-imaging clinics, who must mentally reconcile mammographic, sonographic, and clinical-history data under time pressure, and for patients whose dense tissue or ambiguous BI-RADS category makes a single-modality read unreliable.

At present, clinicians cope by manually cross-referencing separate mammography and ultrasound reports alongside patient charts, a process that is time-consuming, inconsistent across readers, and difficult to audit. Recent evidence supports this concern: unimodal deep-learning classifiers trained only on mammograms plateau near an accuracy of 0.80, whereas cross-attention fusion of mammography with clinical features raises this, and a comprehensive review of multimodal fusion in medical imaging confirms that very few existing systems jointly integrate three or more modalities together with calibrated uncertainty and explainability.

Related work each addresses only part of the problem: cross-attention fusion of mammography and clinical metadata improves accuracy but omits ultrasound; a vision-transformer-based fusion network combines ultrasound with clinical variables but neither uses mammography nor a pretrained foundation-model backbone; and mammography-specific vision-language foundation models improve data efficiency for a single modality without fusing across modalities at all. Building on these gaps, this project proposes an Adaptive Cross-Modal Foundation Transformer (ACMFT) that encodes mammography and ultrasound images through pretrained vision foundation encoders, projects structured clinical metadata through a dedicated encoder, aligns all three streams through cross-modal attention fusion, and passes the fused representation through a multi-task prediction head that outputs a malignancy prediction alongside Grad-CAM and attention-based visual explanations and a calibrated uncertainty estimate.

Within this project's timeline, the work is scoped to benign-versus-malignant classification using publicly available mammography and ultrasound datasets, such as CBIS-DDSM, VinDr-Mammo, INbreast, and BUSI, supplemented with structured clinical metadata, together with ablation and explainability evaluation; it explicitly excludes MRI and histopathology fusion, prospective hospital-deployment, and patent filing, which are identified as future extensions beyond this project's scope. Success will be measured by the ACMFT exceeding the best single-modality baseline's accuracy by a clinically meaningful margin on held-out test data, and by having its saliency-based visual explanations judged consistent with established radiological markers of malignancy in an independent qualitative review of correctly and incorrectly classified cases.

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### References (Category specific)

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2. Mammo-CLIP: A vision-language foundation model to enhance data efficiency and robustness in mammography. Ghosh, S., Poynton, C. B., Visweswaran, S., & Batmanghelich, K.(2024)  
[https://doi.org/10.1007/978-3-031-72390-2\\_59](https://doi.org/10.1007/978-3-031-72390-2_59)
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<https://doi.org/10.1016/j.combiomed.2024.108635>

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